

NEESHA KISHORE BANGERA

AI/ML Engineer

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Available for AI/ML Roles & Internships • Open to Join Immediately

SUMMARY

National Winner, Smart India Hackathon 2025 — selected from 1,000+ competing teams across India. MCA candidate (2026) with proven experience in building AI/ML pipelines, signal processing systems, and data-driven diagnostic tools. Skilled in end-to-end ML workflows — from raw data preprocessing and feature engineering to model deployment and explainable AI visualizations. Passionate about developing intelligent systems that solve real-world engineering challenges.

TECHNICAL SKILLS

Machine Learning: Feature Engineering · Exploratory Data Analysis · Data Preprocessing · Hyperparameter Tuning · Anomaly Detection · AI Agents

NLP: NLTK · spaCy · Text Classification · TF-IDF Vectorization · Sentiment Analysis · Named Entity Recognition (NER) · Tokenization

LLM & Agents: LangChain · LangGraph · RAG (Retrieval-Augmented Generation)

Python & Data: Python · scikit-learn · Pandas · NumPy · Matplotlib · Seaborn · SciPy · ReportLab

Databases: MySQL · Firebase · MongoDB

Tools & Dev: Git · Docker · FastAPI · MLOps

PROJECTS

Vidhyut — AI-Powered Transformer Fault Diagnostics Platform

Smart India Hackathon 2025 —
NATIONAL WINNER

Python · Python · Scikit-learn · SciPy · Matplotlib · CustomTkinter · CSV/XML · Local LLM (RAG)

- Built an ML-based fault detection system for power transformers using Sweep Frequency Response Analysis (SFRA) data, validated on real MSETCL and BHEL datasets across 6 fault categories.
- Reduced manual diagnostic effort by automating multi-format data ingestion (CSV, XML) through a vendor-agnostic preprocessing pipeline that normalized raw FRA signals for ML model input.
- Improved diagnostic transparency by generating explainable AI visualizations that show field engineers exactly why the model flagged a specific fault, without requiring domain expertise.
- Enabled faster, offline-capable maintenance decisions by integrating a local LLM-based AI assistant (RAG) and automating IEC/IEEE-style PDF report generation with cloud storage dispatch.

Automated Smart Wheelchair System

AI | Assistive Technology

Python · Signal Processing · React.js · Firebase

- Developed an AI-powered smart wheelchair integrating EEG brain-signal classification and voice control, enabling hands-free navigation through real-time intent recognition with low-latency response.
- Designed a signal processing pipeline to decode EEG input signals into movement commands, combined with a React.js caregiver dashboard connected to Firebase Realtime Database for live monitoring.
- Applied human-centered design principles to ensure the interface is accessible and usable by individuals with limited physical mobility.

Sentiment Analysis & Text Classification Pipeline

NLP | Python — In Progress

Python · NLTK · scikit-learn · TF-IDF · Flask · spaCy

- Building an end-to-end NLP pipeline using Python, NLTK, and scikit-learn to classify 50K IMDB reviews as positive, negative, or neutral using TF-IDF vectorization with Logistic Regression and Naive Bayes classifiers.
- Evaluating models using precision, recall, and F1-score metrics to confirm generalization on unseen data; planning extension with Named Entity Recognition (NER) using spaCy.
- Designing a Flask REST API for real-time inference to make the classifier accessible as a deployable web service.

EDUCATION

Master of Computer Applications (MCA)

NCRD Sterling Institute of Management Studies, Navi Mumbai

CGPA: 8.62 / 10.00

Aug 2024 – May 2026 (Pursuing)

Bachelor of Computer Applications (BCA)

SMT. P. N. Doshi Women's College, Ghatkopar, Mumbai

CGPA: 8.52 / 10.00

2021 – 2024

ACHIEVEMENTS

- National Winner — Smart India Hackathon 2025 | Selected from 1,000+ teams across India | AI-Powered Transformer Diagnostics (Vidhyut)